

1. Introduction to Cyber Security

Cyber Security Basic For Newbies

Cyber Security is the practice of protecting computers, networks, servers, applications, and data from unauthorized access, attacks, damage, or theft. The goal is to ensure that information remains secure and available only to authorized users.

Simple Definition

Cyber Security is the protection of digital systems and information from cyber threats and attacks.

Real-Life Example

When you use:

- Gmail
- Online Banking
- UPI Payments
- Social Media
- E-commerce Websites

Cyber Security protects your:

- Passwords
- Personal Information
- Bank Details
- Photos and Documents

Without Cyber Security, attackers could steal, modify, or destroy your data.

Why is Cyber Security Important?

Today almost everything is connected to the Internet:

- Mobile Phones
- Laptops
- Smart TVs
- Banking Systems
- Hospitals
- Government Services
- Cloud Applications

This connectivity creates opportunities for cyber criminals.

Common Threats

1. Data Theft

Attackers steal:

- Usernames
- Passwords
- Credit Card Details
- Personal Information

2. Malware Attacks

Malicious software such as:

- Viruses
- Worms
- Trojans
- Ransomware

3. Phishing Attacks

Fake emails or websites designed to steal credentials.

4. Financial Fraud

Attackers may:

- Transfer money
- Perform unauthorized transactions
- Steal banking information

5. Service Disruption

Attacks can make websites or systems unavailable.

Goals of Cyber Security

Cyber Security aims to:

1. Protect Confidentiality

Only authorized users can access sensitive information.

Example: Only you should access your Gmail account.

2. Maintain Integrity

Data should not be modified without authorization.

Example: Marks stored in a university database should not be changed illegally.

3. Ensure Availability

Systems and services should remain accessible when needed.

Example: Online banking should be available 24/7.

These three principles are known as the **CIA Triad**, which is discussed later in the PDF.

Areas of Cyber Security

Network Security

Protecting networks from attacks.

Application Security

Protecting software and web applications.

Information Security

Protecting sensitive data.

Cloud Security

Protecting cloud-based systems and services.

Endpoint Security

Protecting devices such as:

- Computers
- Laptops
- Smartphones

Module 1: Networking Fundamentals

Networking is the foundation of Cyber Security. Before understanding cyber attacks, security controls, and network monitoring, it is important to understand how computers communicate with each other. Networking enables devices to exchange information, share resources, and access services over local and global networks.

2. What is a Network?

A network is a collection of two or more devices connected together for communication and resource sharing. These devices can include computers, servers, printers, smartphones, and networking equipment such as routers and switches.

Subtopics

Definition of Network

A network is a group of interconnected devices that can communicate and exchange data with each other.

Resource Sharing

Networks allow users to share resources such as files, printers, applications, and internet connections.

Device Communication

Devices communicate using protocols such as TCP/IP, enabling reliable data exchange between systems.

Advantages of Networks

- Fast communication
- Resource sharing
- Centralized management
- Cost reduction
- Improved collaboration

3. What is the Internet?

The Internet is the world's largest network that connects millions of computers and networks globally. It enables communication, information sharing, online services, and access to resources from anywhere in the world.

Subtopics

Network of Networks

The Internet is often called a "network of networks" because it connects multiple independent networks into a single global system.

Interconnected Networks

Different organizations, service providers, and countries connect their networks together to form the Internet.

Uses of the Internet

- Email communication
- Web browsing
- Online banking
- Social media
- Cloud computing
- E-commerce

4. How Internet Works?

The Internet works by transmitting data between devices using communication protocols. Every device connected to the Internet has a unique IP address that helps identify and locate it on the network.

Subtopics

ISP (Internet Service Provider)

An ISP provides users with access to the Internet. Examples include Airtel, Jio, BSNL, and ACT.

IP Address

An IP address is a unique identifier assigned to every device connected to a network.

Data Communication Process

When a user requests a website, the request is sent through the network to the destination server.

Data Packets

Large amounts of data are divided into smaller packets before transmission.

Routing

Routers determine the best path for packets to travel from source to destination.

Packet Transmission

Packets travel through multiple networks until they reach their destination.

Packet Reassembly

The destination device collects and reassembles all packets into the original data.

Internet Communication Flow

1. User enters a website URL.
2. DNS resolves the domain name to an IP address.
3. Request is sent to the web server.
4. Server processes the request.
5. Response is sent back to the user.
6. Browser displays the webpage.

5. Types of Networks

Networks are classified based on their size, coverage area, and purpose.

5.1 LAN (Local Area Network)

A LAN connects devices within a small geographical area such as a home, office, school, or building.

Characteristics

- High speed
- Low cost
- Easy maintenance
- Limited coverage area

Examples

- Office network
- School computer lab
- Home Wi-Fi network

5.2 PAN (Personal Area Network)

A PAN is a small network used for connecting personal devices within a short distance.

Characteristics

- Covers a few meters
- Used by a single person
- Usually wireless

Examples

- Bluetooth connection
- Smartwatch connected to a smartphone
- Wireless headphones

5.3 MAN (Metropolitan Area Network)

A MAN connects multiple LANs within a city or metropolitan region.

Characteristics

- Larger than LAN
- Covers a city or town
- High-speed connectivity

Examples

- City-wide government network
- University campus network

5.4 WAN (Wide Area Network)

A WAN covers large geographical areas such as states, countries, or continents.

Characteristics

- Very large coverage area
- Connects multiple LANs and MANs
- Uses public and private communication links

Examples

- The Internet
- Corporate branch office networks
- Banking networks

Module 2: Network Models

Network models provide a structured framework for understanding how data is transmitted between devices across a network. They divide the communication process into multiple layers, with each layer performing specific functions. The two most important network models are the **OSI Model** and the **TCP/IP Model**.

6. OSI Model (Open Systems Interconnection Model)

The OSI Model is a conceptual framework developed by the International Organization for Standardization (ISO) to standardize network communication. It consists of **7 layers**, where each layer performs a specific task in the communication process.

The main purpose of the OSI model is to help understand, design, and troubleshoot network communications.

Layers of the OSI Model

6.1 Application Layer (Layer 7)

The Application Layer is the topmost layer of the OSI model. It provides network services directly to end users and applications.

Protocols

- HTTP (HyperText Transfer Protocol)
- FTP (File Transfer Protocol)
- SMTP (Simple Mail Transfer Protocol)
- DNS (Domain Name System)

Functions

- Web browsing
- Email communication
- File transfer
- Name resolution

Examples

- Accessing websites through a browser
- Sending emails
- Downloading files

6.2 Presentation Layer (Layer 6)

The Presentation Layer ensures that data is presented in a format understandable by both sender and receiver.

Functions

Data Conversion

Converts data into a standard format.

Data Compression

Reduces data size to improve transmission efficiency.

Encryption

Secures sensitive information during transmission.

Examples

- SSL/TLS encryption
- JPEG image formatting
- MP3 audio compression

6.3 Session Layer (Layer 5)

The Session Layer establishes, manages, and terminates communication sessions between devices.

Functions

Session Establishment

Creates a communication session between devices.

Authentication

Verifies the identity of users or systems.

Authorization

Determines access permissions and privileges.

Examples

- User login sessions
- Video conferencing connections
- Remote desktop sessions

6.4 Transport Layer (Layer 4)

The Transport Layer ensures reliable and efficient delivery of data between devices.

Protocols

TCP (Transmission Control Protocol)

- Reliable communication
- Error checking
- Connection-oriented

UDP (User Datagram Protocol)

- Faster communication
- Connectionless
- Used for real-time applications

Functions

Segmentation

Breaks large data into smaller segments.

Flow Control

Controls the rate of data transmission.

Error Control

Detects and corrects transmission errors.

Examples

- TCP: Web browsing, email
- UDP: Online gaming, video streaming

6.5 Network Layer (Layer 3)

The Network Layer is responsible for routing packets between different networks.

Functions

Logical Addressing

Uses IP addresses to identify devices.

Routing

Determines the best path for packet delivery.

Path Determination

Selects the most efficient route to the destination.

Examples

- Routers
- IPv4 and IPv6 addressing

6.6 Data Link Layer (Layer 2)

The Data Link Layer provides node-to-node communication and handles physical addressing.

Functions

Physical Addressing

Uses MAC addresses to identify devices on a local network.

MAC Addressing

Ensures correct delivery of frames within the same network.

Examples

- Switches
- Ethernet communication
- MAC Address management

6.7 Physical Layer (Layer 1)

The Physical Layer is responsible for transmitting raw bits over a physical medium.

Components

Cables

Transmit electrical or optical signals.

Bridges

Connect multiple network segments.

NIC (Network Interface Card)

Allows devices to connect to networks.

Hardware Connectivity

Includes connectors, hubs, repeaters, and transmission media.

Examples

- Ethernet cables
- Fiber optic cables
- Wireless transmission signals

Data Flow in the OSI Model

When data is sent:

1. Application Layer creates data.
2. Presentation Layer formats and encrypts data.
3. Session Layer establishes communication.
4. Transport Layer segments data.
5. Network Layer adds IP addresses.
6. Data Link Layer adds MAC addresses.
7. Physical Layer transmits bits through the medium.

7. TCP/IP Model

The TCP/IP Model is the practical networking model used on the Internet today. It is simpler than the OSI model and focuses on real-world communication.

The TCP/IP model defines how data should be transmitted between devices over networks.

TCP/IP Architecture

The TCP/IP model organizes network communication into layers.

Main Goals

- Reliable communication
- Interoperability
- Scalability
- Efficient data transfer

4-Layer TCP/IP Model

1. Application Layer

Combines the functions of:

- Application Layer
- Presentation Layer
- Session Layer

Protocols:

- HTTP
- HTTPS
- FTP
- SMTP
- DNS

2. Transport Layer

Protocols:

- TCP
- UDP

Functions:

- Reliable communication
- Segmentation
- Flow control

3. Internet Layer

Functions:

- IP addressing
- Routing
- Packet forwarding

Protocols:

- IPv4
- IPv6
- ICMP

4. Network Access Layer

Functions:

- Physical transmission

- MAC addressing
- Frame delivery

5-Layer TCP/IP Model

Many modern networking books describe TCP/IP using a 5-layer structure:

1. Application Layer
2. Transport Layer
3. Network Layer
4. Data Link Layer
5. Physical Layer

This structure closely resembles the OSI model.

OSI Model vs TCP/IP Model

Feature	OSI Model	TCP/IP Model
Layers	7	4 or 5
Developed By	ISO	DARPA
Usage	Reference Model	Practical Model
Complexity	More Detailed	Simpler
Real-world Use	Educational	Internet Communication

Encapsulation

Encapsulation is the process of adding protocol information at each layer before data is transmitted over a network.

Data Encapsulation Process

As data moves from the sender to the receiver:

1. Application Layer creates data.
2. Transport Layer adds TCP/UDP header.
3. Network Layer adds IP header.
4. Data Link Layer adds MAC header and trailer.
5. Physical Layer converts data into signals.

Layer-wise Data Addition

Layer	Data Unit
-------	-----------

Application	Data
-------------	------

Transport	Segment
-----------	---------

Network	Packet
---------	--------

Data Link	Frame
-----------	-------

Physical	Bits
----------	------

Example

When you open a website:

- Browser creates an HTTP request.
- TCP adds transport information.
- IP adds source and destination addresses.
- Ethernet adds MAC addresses.
- Data is transmitted as bits through the network.

This process is called **Encapsulation**, and the reverse process at the receiver side is called **Decapsulation**.

Importance in Cyber Security

Understanding OSI and TCP/IP models helps security professionals:

- Analyze network traffic
- Detect attacks
- Perform packet analysis
- Troubleshoot network issues
- Understand firewall operations
- Investigate security incidents

Module 3: Transport Layer Protocols

The Transport Layer is the fourth layer of the OSI Model and is responsible for end-to-end communication between devices. It ensures that data is delivered correctly, efficiently, and reliably from the sender to the receiver.

The two main protocols used at the Transport Layer are:

- TCP (Transmission Control Protocol)

- UDP (User Datagram Protocol)

Both protocols serve different purposes depending on the application's requirements.

8. TCP vs UDP

TCP and UDP are communication protocols that operate at the Transport Layer. The major difference between them is reliability.

Feature	TCP	UDP
Connection Type	Connection-Oriented	Connectionless
Reliability	Reliable	Unreliable
Error Checking	Yes	Limited
Speed	Slower	Faster
Data Ordering	Maintained	Not Guaranteed
Overhead	High	Low
Usage	Web, Email, File Transfer	Gaming, Streaming, VoIP

TCP (Transmission Control Protocol)

TCP is a connection-oriented protocol that provides reliable communication between devices. It ensures that all data packets reach their destination correctly and in the proper order.

TCP is widely used when data accuracy and reliability are important.

Examples

- Web Browsing (HTTP/HTTPS)
- Email Services
- File Transfer
- Online Banking
- Database Communication

Reliable Communication

TCP establishes a connection before transmitting data and verifies that data reaches the destination successfully.

Features of Reliable Communication:

- Error Detection
- Error Recovery
- Packet Reordering
- Data Integrity
- Flow Control

If a packet is lost during transmission, TCP automatically retransmits it.

3-Way Handshake

Before communication begins, TCP establishes a connection using a process called the **Three-Way Handshake**.

Step 1: SYN

The client sends a SYN (Synchronize) packet to the server requesting a connection.

Client ----- SYN -----> Server

Step 2: SYN-ACK

The server responds with a SYN-ACK packet indicating that it received the request and is ready to communicate.

Client <---- SYN-ACK ---- Server

Step 3: ACK

The client sends an ACK packet confirming the connection.

Client ----- ACK -----> Server

After these three steps, the connection is established and data transfer begins.

Acknowledgement Number

An Acknowledgement Number confirms that a packet has been received successfully.

Purpose

- Confirms packet delivery
- Detects missing packets
- Enables retransmission if necessary

Example

Packet 1 Sent

ACK Received

Packet 2 Sent

ACK Received

If ACK is not received, TCP resends the packet.

Sequence Number

Each TCP segment is assigned a unique sequence number.

Purpose

- Maintains packet order
- Reassembles data correctly
- Detects missing packets

Example

Segment 1

Segment 2

Segment 3

Segment 4

Even if packets arrive out of order, the receiver uses sequence numbers to arrange them correctly.

Checksum

A Checksum is used to verify data integrity during transmission.

Purpose

- Detect transmission errors
- Ensure data accuracy
- Verify packet contents

Working

1. Sender calculates checksum.
2. Receiver calculates checksum again.
3. Both values are compared.
4. If values match, data is accepted.
5. If values differ, packet is discarded and retransmitted.

UDP (User Datagram Protocol)

UDP is a connectionless protocol that sends data without establishing a connection first.

Unlike TCP, UDP does not guarantee delivery, ordering, or error recovery.

Its primary advantage is speed.

Unreliable Communication

UDP sends packets without:

- Acknowledgements
- Sequence Numbers
- Retransmissions
- Connection Establishment

Because of this, some packets may be lost during transmission.

However, the communication is significantly faster than TCP.

Real-Time Communication

UDP is ideal for real-time applications where speed is more important than perfect accuracy.

Examples include:

- Video Conferencing
- Live Streaming
- Online Gaming
- Voice Calls

A slight loss of data is usually acceptable in these applications.

Gaming

Most multiplayer online games use UDP because:

- Low latency is required.
- Fast response times are important.
- Missing a few packets is less harmful than delays.

Examples

- PUBG
- Call of Duty
- Valorant
- Fortnite

Voice Calls

Voice-over-IP (VoIP) services use UDP because real-time audio transmission is critical.

Examples:

- WhatsApp Calls
- Zoom Audio
- Skype Calls
- Discord Voice Chat

It is better to lose a few audio packets than to experience delays.

Streaming

Video and audio streaming applications use UDP to provide smooth playback.

Examples:

- YouTube Live
- Netflix Streaming
- Twitch
- IPTV Services

Minor packet loss usually goes unnoticed by users.

TCP vs UDP Use Cases

Use TCP When:

- Data accuracy is important.
- Files must not be corrupted.
- Transactions must be reliable.

Examples:

- Banking Applications
- Email Systems
- Websites
- File Downloads

Use UDP When:

- Speed is critical.
- Real-time communication is required.
- Small packet losses are acceptable.

Examples:

- Online Games
- Live Video Streaming
- Voice Calls
- Video Conferencing

Importance in Cyber Security

Understanding TCP and UDP is essential for:

- Network Traffic Analysis
- Packet Capture Analysis
- Firewall Configuration
- Intrusion Detection Systems (IDS)
- Intrusion Prevention Systems (IPS)
- Security Monitoring
- Incident Response

Module 4: IP Addressing

IP Addressing is one of the most important concepts in networking and cyber security. Every device connected to a network requires a unique address to communicate with other devices. This address is known as an Internet Protocol (IP) Address.

IP addresses help identify devices, route data packets, and ensure communication occurs between the correct source and destination systems.

9. What is an IP Address?

An IP Address (Internet Protocol Address) is a unique numerical identifier assigned to every device connected to a network. It enables devices to communicate with each other and exchange data over the Internet or private networks.

Just as a home address helps deliver a letter to the correct location, an IP address helps deliver data packets to the correct device.

Internet Protocol (IP)

Internet Protocol is a set of rules that governs how data is transmitted across networks.

The protocol is responsible for:

- Addressing devices
- Routing packets
- Identifying source and destination systems

- Enabling communication between networks

Every packet transmitted over the Internet contains source and destination IP addresses.

Device Identification

Every device connected to a network requires a unique identifier.

Examples of devices that use IP addresses:

- Computers
- Laptops
- Smartphones
- Servers
- Routers
- Printers
- IoT Devices

Without IP addresses, devices would not know where to send or receive data.

TCP/IP Communication

The TCP/IP protocol suite uses IP addresses to establish communication between devices.

Communication Process

1. Sender creates data.
2. TCP divides data into segments.
3. IP adds source and destination addresses.
4. Packets travel through routers.
5. Destination device receives packets.
6. TCP reassembles the original data.

Example:

Source Device (192.168.1.10)



Router



Internet Network



Server

(142.250.183.14)

10. Types of IP Addresses

There are different versions of IP addresses used in networking.

Types

- IPv4 (Internet Protocol Version 4)
- IPv6 (Internet Protocol Version 6)
- IPv5 (Experimental/Theoretical Version)

IPv4 (Internet Protocol Version 4)

IPv4 is the most commonly used version of IP addressing.

It uses a **32-bit address structure**, allowing approximately **4.3 billion unique addresses**.

Example

192.168.1.1

8.8.8.8

172.16.0.1

32-bit Addressing

IPv4 consists of 32 bits divided into four equal sections.

Each section contains 8 bits.

11000000.10101000.00000001.00000001

The binary value above represents:

192.168.1.1

Octets

Each 8-bit section of an IPv4 address is called an **Octet**.

Example:

192.168.1.1

Octet 1 Octet 2 Octet 3 Octet 4

192 168 1 1

Each octet can have values from:

0 to 255

Address Structure

IPv4 addresses contain two parts:

Network Portion

Identifies the network.

Host Portion

Identifies the device within the network.

Example:

192.168.1.25

- Network = 192.168.1
- Host = 25

This structure helps routers forward packets efficiently.

IPv6 (Internet Protocol Version 6)

IPv6 was developed to overcome the address limitations of IPv4.

It uses a **128-bit address structure**, providing an enormous number of unique addresses.

Example

2001:0db8:85a3:0000:0000:8a2e:0370:7334

128-bit Addressing

IPv6 uses 128 bits compared to IPv4's 32 bits.

This allows approximately:

340 undecillion addresses

Which is:

340,282,366,920,938,463,463,374,607,431,768,211,456

addresses.

Advantages of IPv6

- Larger address space
- Better security support
- Improved routing
- Reduced network congestion
- Efficient packet processing

Transition Mechanisms

Organizations cannot instantly switch from IPv4 to IPv6.

Several transition techniques are used.

Dual Stacking

A device runs both IPv4 and IPv6 simultaneously.

Example:

IPv4 : 192.168.1.10

IPv6 : 2001:db8::1

Benefits:

- Smooth migration
- Supports both protocols

Tunneling

IPv6 packets travel through IPv4 networks using tunnels.

This allows IPv6 communication even when intermediate networks only support IPv4.

Benefits:

- Compatibility
- Gradual migration

NAT (Network Address Translation)

NAT translates addresses between different network protocols.

Functions:

- Conserves IP addresses
- Enables communication between networks
- Supports transition strategies

Example:

Private IP → Public IP

IPv5

IPv5 was an experimental version of the Internet Protocol.

It was developed primarily for:

- Multimedia streaming
- Voice communication

However, IPv5 was never officially deployed on the Internet because IPv6 was already being developed.

Therefore, IPv5 is considered a theoretical or experimental version.

11. Classes of IPv4 Addresses

IPv4 addresses are divided into different classes based on network size and usage.

Class A

Used for very large networks.

Range

1.0.0.0 to 126.255.255.255

Characteristics

- Large number of hosts
- Few networks
- Used by large organizations

Example

10.0.0.1

Class B

Used for medium-sized networks.

Range

128.0.0.0 to 191.255.255.255

Characteristics

- Moderate number of hosts
- Common in universities and enterprises

Example

172.16.0.1

Class C

Used for small networks.

Range

192.0.0.0 to 223.255.255.255

Characteristics

- Large number of networks
- Fewer hosts per network

Example

192.168.1.1

Class D

Reserved for Multicast communication.

Range

224.0.0.0 to 239.255.255.255

Usage

- Video broadcasting
- Online streaming
- Group communication

Class E

Reserved for research and experimental purposes.

Range

240.0.0.0 to 255.255.255.255

Usage

- Future development
- Experimental networking

Summary Table of IPv4 Classes

Class	Range	Purpose
A	1 – 126	Large Networks
B	128 – 191	Medium Networks
C	192 – 223	Small Networks
D	224 – 239	Multicasting
E	240 – 255	Experimental

Importance in Cyber Security

Understanding IP addressing helps security professionals:

- Track attackers
- Analyze network traffic
- Configure firewalls

- Detect suspicious connections
- Investigate security incidents
- Perform penetration testing
- Conduct vulnerability assessments

Module 5: Ports and Browser Working

Ports and browsers play a crucial role in network communication and web browsing. Whenever a user accesses a website, sends an email, downloads a file, or uses an online application, network ports and web browsers work together to establish communication between the client and the server.

Understanding ports and browser operations is important for networking, web development, penetration testing, and cyber security.

12. Port Numbers

A Port Number is a logical communication endpoint used by applications and services running on a computer. While an IP address identifies a device on a network, a port number identifies a specific service or application running on that device.

Think of it this way:

- IP Address = Building Address
- Port Number = Room Number inside the building

For example:

192.168.1.10:80

- IP Address → 192.168.1.10
- Port Number → 80 (HTTP Service)

This helps the operating system know which application should receive incoming data.

Why Are Ports Important?

Ports allow multiple services to run simultaneously on a single machine.

For example, a server can provide:

- Website on Port 80
- Secure Website on Port 443
- Email Service on Port 25
- DNS Service on Port 53

All using the same IP address.

Port Ranges

Port numbers range from:

0 – 65535

These ports are divided into three categories.

Well-Known Ports (0 – 1023)

These ports are reserved for commonly used services and protocols.

Characteristics

- Standardized ports
- Used by system services
- Recognized worldwide

Examples

Port	Protocol	Purpose
------	----------	---------

20,21	FTP	File Transfer
23	TELNET	Remote Login
25	SMTP	Email Sending
53	DNS	Domain Name Resolution
80	HTTP	Web Browsing
143	IMAP	Email Retrieval
443	HTTPS	Secure Web Browsing

Registered Ports (1024 – 49151)

These ports are assigned to specific applications and software vendors.

Characteristics

- Used by user applications
- Registered with IANA
- Commonly used by enterprise software

Examples

- Database Services

- Application Servers
- Business Applications

Dynamic/Private Ports (49152 – 65535)

These ports are temporarily assigned by the operating system when a client initiates a connection.

Characteristics

- Short-lived
- Automatically assigned
- Used for client-side communication

Example

When you open a website:

Your Computer:

192.168.1.5:53001

Website:

142.250.183.14:443

Port 53001 is dynamically assigned by the operating system.

Common Ports

FTP (Ports 20, 21)

FTP (File Transfer Protocol) is used for transferring files between computers.

Uses

- Uploading website files
- Downloading documents
- File sharing

TELNET (Port 23)

TELNET is a protocol used for remote access to systems.

Features

- Command-line access
- Remote administration

Security Note

TELNET transmits data in plain text and is considered insecure today.

SSH is commonly used instead.

SMTP (Port 25)

Simple Mail Transfer Protocol

Used for sending emails between mail servers.

Functions

- Email delivery
- Email forwarding
- Mail routing

DNS (Port 53)

Domain Name System

Converts domain names into IP addresses.

Example

google.com

↓

142.250.183.14

Without DNS, users would have to remember IP addresses instead of domain names.

HTTP (Port 80)

HyperText Transfer Protocol

Used for transferring web pages between browsers and web servers.

Characteristics

- Unencrypted communication
- Used for standard websites

Example:

http://example.com

IMAP (Port 143)

Internet Message Access Protocol

Used for retrieving and managing emails from mail servers.

Features

- Synchronizes emails
- Supports multiple devices

HTTPS (Port 443)

HyperText Transfer Protocol Secure

A secure version of HTTP that encrypts communication using SSL/TLS.

Benefits

- Data encryption
- Confidential communication
- Protection against interception

Example:

<https://bank.com>

Why Ports Matter in Cyber Security

Security professionals frequently analyze open ports because attackers often target vulnerable services.

Activities

- Port Scanning
- Vulnerability Assessment
- Service Enumeration
- Firewall Configuration
- Intrusion Detection

Common tools:

- Nmap
- Netcat
- Wireshark
- Nessus

13. How Browser Works?

A Web Browser is a software application that allows users to access websites and web applications over the Internet.

Popular browsers include:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

- Safari
- Opera

When a user enters a URL, the browser performs several operations before displaying the webpage.

URL Processing

The process begins when a user enters a URL into the browser.

Example:

`https://www.google.com`

The browser analyzes the URL and identifies:

- Protocol (HTTPS)
- Domain Name (google.com)
- Resource Path (if any)

DNS Resolution

The browser must find the IP address associated with the domain name.

Process

1. Browser checks cache.
2. Operating system checks cache.
3. DNS server is queried.
4. DNS returns IP address.

Example:

`google.com`

↓

`142.250.183.14`

This process is called DNS Resolution.

Request Handling

Once the IP address is obtained:

Step 1

Browser establishes a TCP connection.

Step 2

HTTPS websites perform SSL/TLS handshake.

Step 3

Browser sends an HTTP request.

Example:

GET / HTTP/1.1

Host: www.google.com

Step 4

Server processes the request.

Step 5

Server sends a response.

Example:

HTTP/1.1 200 OK

The response contains webpage resources.

HTML/CSS/JS Rendering

After receiving the response, the browser renders the webpage.

HTML Parsing

HTML creates the webpage structure.

Example:

```
<h1>Welcome</h1>
```

CSS Processing

CSS controls the appearance of the webpage.

Example:

```
h1 {  
  color: blue;  
}
```

JavaScript Execution

JavaScript adds interactivity and dynamic behavior.

Example:

```
alert("Hello");
```

Rendering Process

1. HTML is parsed.

2. CSS is applied.
3. JavaScript executes.
4. DOM is created.
5. Browser renders the page.

Complete Browser Workflow

User enters URL



URL Processing



DNS Resolution



TCP Connection



HTTP/HTTPS Request



Server Processing



HTTP Response



HTML/CSS/JS Download



Browser Rendering



Webpage Displayed

Importance in Cyber Security

Understanding browser operations helps security professionals identify:

- Cross-Site Scripting (XSS)
- Session Hijacking
- Man-in-the-Middle Attacks
- DNS Spoofing
- Phishing Websites
- SSL/TLS Issues
- Web Application Vulnerabilities

Module 6: Web Security Fundamentals

Web Security is a branch of Cyber Security that focuses on protecting websites, web applications, servers, and users from cyber threats. Since most online activities such as banking, shopping, email, and social media are performed through web applications, understanding web security concepts is essential.

The foundation of web security begins with understanding HTTP, HTTPS, SSL, TLS, cryptography, and URL structures.

14. HTTP vs HTTPS

HTTP and HTTPS are communication protocols used between a web browser and a web server. They define how information is exchanged over the Internet.

What is HTTP?

HTTP (HyperText Transfer Protocol) is the standard protocol used for transmitting web pages and data over the Internet.

Characteristics

- Transfers data in plain text
- No encryption
- Faster but less secure
- Uses Port 80

Example

`http://example.com`

Risks of HTTP

Since data is transmitted in plain text:

- Passwords can be intercepted
- Sensitive information can be stolen
- Vulnerable to Man-in-the-Middle (MITM) attacks

What is HTTPS?

HTTPS (HyperText Transfer Protocol Secure) is the secure version of HTTP.

It uses SSL/TLS encryption to protect communication between the browser and server.

Characteristics

- Encrypted communication
- Secure data transmission

- Authentication of websites
- Uses Port 443

Example

https://example.com

Benefits

- Protects user credentials
- Prevents data interception
- Ensures confidentiality
- Increases user trust

Secure Communication

Secure communication ensures that information exchanged between a client and server remains:

Confidential

Only authorized parties can view the data.

Authentic

Users can verify the identity of the website.

Untampered

Data cannot be modified during transmission.

Example

When logging into an online banking website:

User → HTTPS → Bank Server

The communication remains protected from attackers.

Encryption

Encryption is the process of converting readable information (plaintext) into an unreadable format (ciphertext).

Purpose

- Protect sensitive information
- Prevent unauthorized access
- Secure communication channels

Example

Plain Text:
Password123

Encrypted Text:
A8x\$9P!kL2#

Only authorized users with the correct key can decrypt the information.

SSL (Secure Sockets Layer)

SSL is a security protocol developed to provide secure communication over the Internet.

Functions

- Encrypts data
- Authenticates websites
- Protects user information

Although SSL is largely obsolete today, the term is still commonly used.

TLS (Transport Layer Security)

TLS is the modern successor to SSL.

Advantages

- Stronger encryption
- Improved performance
- Better security mechanisms
- Protection against modern attacks

Common TLS Versions

- TLS 1.2
- TLS 1.3

Today, most websites use TLS rather than SSL.

HTTP vs HTTPS Comparison

Feature	HTTP HTTPS	
Security	No	Yes
Encryption	No	SSL/TLS
Port	80	443

Feature	HTTP	HTTPS
---------	------	-------

Data Protection	No	Yes
-----------------	----	-----

Authentication	No	Yes
----------------	----	-----

Browser Trust	Low	High
---------------	-----	------

15. SSL and TLS

SSL and TLS are cryptographic protocols that secure communication between browsers and servers.

They rely on two important cryptographic techniques:

- Symmetric Cryptography
- Asymmetric Cryptography

Both work together during HTTPS communication.

Symmetric Cryptography

Symmetric Cryptography uses the same key for both encryption and decryption.

Same Key Encryption

A single shared key is used by both sender and receiver.

Process

Plain Text

↓

Encryption Key

↓

Cipher Text

↓

Same Key

↓

Original Data

Example

If the key is:

KEY123

The same key is used for:

- Encrypting data
- Decrypting data

Advantages

- Fast encryption
- Efficient processing
- Suitable for large amounts of data

Disadvantages

- Key distribution is difficult
- If the key is stolen, communication is compromised

Common Symmetric Algorithms

- AES (Advanced Encryption Standard)
- DES
- 3DES
- Blowfish

Decryption

Decryption is the process of converting encrypted data back into readable form using the same secret key.

Example

Encrypted Data



Secret Key



Original Data

Only users possessing the key can access the information.

Asymmetric Cryptography

Asymmetric Cryptography uses two mathematically related keys.

- Public Key
- Private Key

This method is also known as Public Key Cryptography.

Public Key

The Public Key can be shared openly with anyone.

Purpose

- Encrypt data

- Verify digital signatures

Example

A website publishes its public key inside its SSL/TLS certificate.

Private Key

The Private Key remains secret and is stored securely on the server.

Purpose

- Decrypt encrypted data
- Create digital signatures

Security Rule

Public Key → Shared

Private Key → Secret

Public Key Encryption

Process

1. Sender obtains receiver's Public Key.
2. Data is encrypted using the Public Key.
3. Encrypted data is sent.
4. Receiver uses Private Key to decrypt.

Example

Message

↓

Public Key

↓

Encrypted Message

↓

Private Key

↓

Original Message

Only the owner of the Private Key can decrypt the information.

Why HTTPS Uses Both Methods

HTTPS combines both encryption methods:

Step 1

Asymmetric Cryptography is used to securely exchange keys.

Step 2

Symmetric Cryptography is used for actual communication because it is faster.

This hybrid approach provides both security and performance.

16. URL (Uniform Resource Locator)

A URL is the address used to locate resources on the Internet.

It tells the browser:

- Which protocol to use
- Which server to contact
- Which resource to retrieve

Example URL

<https://accounts.google.com/user/mail/index.php?id=101#profile>

URL Components

1. Scheme

The scheme specifies the protocol used for communication.

Examples

http
https
ftp
smtp

Example:

<https://example.com>

Here:

https

is the scheme.

2. Subdomain

A subdomain is a subdivision of the main domain.

Examples

www.google.com
mail.google.com
accounts.google.com
docs.google.com

Here:

accounts

is the subdomain.

3. Domain

The domain identifies the website.

Example

google

in:

google.com

The domain is easier to remember than an IP address.

4. Top-Level Domain (TLD)

The TLD is the extension appearing after the domain.

Common TLDs

.com

.org

.net

.edu

.gov

.in

Example:

google.com

Here:

.com

is the TLD.

5. File Path

The file path specifies the exact resource being requested.

Example

https://example.com/products/mobile

File Path:

/products/mobile

6. Parameter String

Parameters provide additional information to the server.

Example

<https://example.com/search?q=laptop&sort=price>

Parameters:

q=laptop
sort=price

Parameters begin after the:

?

symbol.

7. Anchor

An anchor points to a specific section within a webpage.

Example

<https://example.com/page#contact>

Anchor:

#contact

The browser jumps directly to that section of the page.

Complete URL Breakdown

Example:

<https://accounts.google.com/user/mail/index.php?id=101#profile>

Component Value

Scheme https

Subdomain accounts

Domain google

TLD .com

File Path /user/mail/index.php

Parameter id=101

Anchor #profile

Importance in Cyber Security

Understanding HTTP, HTTPS, SSL/TLS, and URLs helps security professionals:

- Detect phishing websites
- Analyze malicious URLs
- Investigate web attacks
- Configure SSL certificates
- Secure web applications
- Identify Man-in-the-Middle attacks
- Understand secure communications

Module 7: HTTP Fundamentals

HTTP (HyperText Transfer Protocol) is the foundation of data communication on the World Wide Web. It is an application-layer protocol used for communication between clients (web browsers) and servers.

Whenever you visit a website, submit a form, log in to an application, or download content, HTTP is used behind the scenes to exchange information between your browser and the web server.

Understanding HTTP is essential for Web Security, Penetration Testing, SOC Analysis, Vulnerability Assessment, and Cyber Security.

17. HTTP Methods

HTTP methods define the action that a client wants to perform on a server resource. They are also known as HTTP Verbs.

Each request sent to a server contains an HTTP method that tells the server what operation should be performed.

GET Method

The GET method is used to retrieve data from a server.

Characteristics

- Fetches information
- Does not modify data
- Parameters are visible in the URL
- Most commonly used method

Example

GET /products HTTP/1.1

Host: example.com

Use Cases

- Opening a webpage
- Viewing products
- Searching content
- Fetching user information

Example URL

<https://example.com/products>

POST Method

The POST method is used to send data to a server.

Characteristics

- Creates new resources
- Data is sent in the request body
- More secure than GET for sensitive data

Example

POST /register HTTP/1.1

Host: example.com

Use Cases

- User registration
- Login forms
- Uploading files
- Creating records

PUT Method

The PUT method is used to update existing resources.

Characteristics

- Modifies existing data
- Replaces the entire resource

Example

PUT /users/100 HTTP/1.1

Use Cases

- Updating profile information
- Editing records
- Modifying application data

DELETE Method

The DELETE method removes resources from a server.

Example

DELETE /users/100 HTTP/1.1

Use Cases

- Deleting user accounts
- Removing files
- Deleting records

HEAD Method

The HEAD method retrieves only the response headers without the response body.

Purpose

- Check server status
- Verify resource availability
- Retrieve metadata

Example

HEAD /index.html HTTP/1.1

CONNECT Method

The CONNECT method establishes a secure connection between a client and a server.

Common Uses

- HTTPS communication
- Proxy servers
- Secure tunnels

OPTIONS Method

The OPTIONS method identifies which HTTP methods are supported by a server.

Example

OPTIONS / HTTP/1.1

Use Cases

- API testing
- Security assessments
- Browser preflight requests

TRACE Method

The TRACE method echoes the received request back to the client.

Purpose

- Debugging
- Request analysis

Security Note

TRACE is often disabled because it can be exploited in Cross-Site Tracing (XST) attacks.

HTTP Methods Summary

Method	Purpose
--------	---------

GET	Retrieve Data
-----	---------------

POST	Create Data
------	-------------

PUT	Update Data
-----	-------------

DELETE	Remove Data
--------	-------------

HEAD	Retrieve Headers
------	------------------

CONNECT	Establish Secure Connection
---------	-----------------------------

OPTIONS	Discover Supported Methods
---------	----------------------------

TRACE	Debug Requests
-------	----------------

18. HTTP Header Fields

HTTP Headers contain additional information about requests and responses. They help clients and servers exchange metadata about communication.

Headers are transmitted as key-value pairs.

Example

Host: example.com
User-Agent: Chrome
Content-Type: application/json

General Headers

General headers apply to both requests and responses.

Examples

Date
Connection
Cache-Control
Pragma

Purpose

- Control caching
- Manage connections
- Provide timestamps

Request Headers

Request headers are sent by the client to the server.

Examples

Host
User-Agent
Cookie
Authorization
Referer
Accept

Functions

- Identify browser
- Send authentication information
- Specify accepted content types

Example

GET / HTTP/1.1
Host: google.com
User-Agent: Chrome

Response Headers

Response headers are sent by the server to the client.

Examples

Server

Set-Cookie

Location

Content-Type

Functions

- Identify server software
- Set cookies
- Redirect users

Example

HTTP/1.1 200 OK

Server: Apache

Entity Headers

Entity headers describe the content being transferred.

Examples

Content-Length

Content-Type

Content-Encoding

Purpose

- Define data type
- Specify content size
- Indicate compression

19. HTTP Request and Response

HTTP communication follows a Request-Response model.

Client

Usually a browser or application.

Server

Processes requests and sends responses.

HTTP Request Structure

An HTTP Request contains information sent from the client to the server.

Header

Headers contain metadata about the request.

Example

Host: example.com

User-Agent: Chrome

Authorization: Bearer Token

Body

The request body contains actual data sent to the server.

Typically used with:

- POST
- PUT
- PATCH

Example

```
{  
  "username":"admin",  
  "password":"12345"  
}
```

Complete Request Example

POST /login HTTP/1.1

Host: example.com

Content-Type: application/json

```
{  
  "username":"admin",  
  "password":"12345"  
}
```

HTTP Response Structure

The server sends a response after processing the request.

Status Line

The status line contains:

- HTTP Version
- Status Code
- Status Message

Example

HTTP/1.1 200 OK

Headers

Response headers provide additional information.

Example

Server: Apache

Content-Type: text/html

Set-Cookie: SessionID=12345

Body

The response body contains actual content.

Examples

- HTML pages
- JSON data
- Images
- Documents

Example

```
<h1>Welcome User</h1>
```

Complete Response Example

HTTP/1.1 200 OK

Content-Type: text/html

```
<html>
```

```
<h1>Welcome</h1>
```

```
</html>
```

HTTP Communication Flow

Browser



HTTP Request



Web Server



HTTP Response



Browser

20. HTTP Status Codes

HTTP Status Codes indicate the result of a client's request.

Every HTTP response contains a status code.

1XX Informational

These codes indicate that the request has been received and processing is continuing.

Examples

Code Meaning

100 Continue

101 Switching Protocols

2XX Success

These codes indicate that the request was successfully processed.

Examples

Code Meaning

200 OK

201 Created

202 Accepted

204 No Content

Most Common

200 OK

3XX Redirection

These codes indicate that the client must take additional action to complete the request.

Examples

Code Meaning

301 Moved Permanently

302 Found

304 Not Modified

Example

301 Moved Permanently

Used when websites redirect users to a new URL.

4XX Client Error

These codes indicate that the client made an incorrect request.

Examples

Code Meaning

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

405 Method Not Allowed

Common Example

404 Not Found

Occurs when the requested page does not exist.

5XX Server Error

These codes indicate a problem on the server side.

Examples

Code Meaning

500 Internal Server Error

502 Bad Gateway

503 Service Unavailable

504 Gateway Timeout

Common Example

500 Internal Server Error

Occurs when the server encounters an unexpected problem.

Importance in Cyber Security

Understanding HTTP fundamentals helps security professionals:

- Analyze web traffic

- Perform penetration testing
- Detect malicious requests
- Investigate web attacks
- Identify authentication issues
- Test APIs
- Monitor suspicious activities

Module 8: Privacy Technologies

Privacy Technologies are tools and techniques used to protect users' identity, data, and online activities while browsing the Internet. They help maintain confidentiality, prevent tracking, and secure communication between users and online services.

Two commonly used privacy technologies are **Proxy Servers** and **VPNs (Virtual Private Networks)**.

21. Proxy

A Proxy Server acts as an intermediary between a user and the Internet. Instead of connecting directly to a website, the user's request first goes to the proxy server, which then forwards the request to the destination server.

The destination server sees the proxy server's IP address rather than the user's actual IP address.

Communication Flow

User → Proxy Server → Website

Website → Proxy Server → User

Anonymous Browsing

One of the primary uses of a proxy server is anonymous browsing.

Benefits

- Hides user identity
- Reduces online tracking
- Prevents websites from seeing the actual IP address

Example

Without Proxy:

User IP → Website

With Proxy:

Proxy IP → Website

The website sees only the proxy server's IP address.

IP Hiding

Every device connected to the Internet has a public IP address.

A proxy server masks this IP address by replacing it with its own.

Advantages

- Improved privacy
- Reduced tracking
- Location masking

Example

Original IP : 192.168.1.100

Visible IP : Proxy Server IP

Proxy Servers

A Proxy Server receives requests from clients and forwards them to the destination server.

Functions

- IP masking
- Content filtering
- Access control
- Caching
- Anonymous browsing

Types of Proxy Servers

Forward Proxy

Represents the client to the Internet.

Reverse Proxy

Represents the server to users.

Transparent Proxy

Users may not know it exists.

Anonymous Proxy

Hides the user's identity.

Advantages of Proxy

- Faster access through caching
- Basic anonymity
- Content filtering
- Access restrictions

Limitations of Proxy

- Usually does not encrypt traffic
- Limited security
- Some proxies log user activity
- Vulnerable to interception

22. VPN (Virtual Private Network)

A VPN creates a secure, encrypted tunnel between a user's device and a VPN server.

Unlike a proxy, a VPN encrypts all network traffic, providing both privacy and security.

Communication Flow

User



Encrypted Tunnel



VPN Server



Internet

Encryption

Encryption is the most important feature of a VPN.

Purpose

- Protect sensitive information
- Prevent eavesdropping
- Secure public Wi-Fi usage

Example

Without VPN:

User → Internet

Data can potentially be intercepted.

With VPN:

User → Encrypted Tunnel → VPN Server

Data remains protected.

Secure Routing

VPNs route user traffic through secure VPN servers.

Benefits

- Improved privacy
- Secure communication
- Protection from network monitoring

Example

Instead of:

User → ISP → Website

VPN Routing:

User → VPN → Website

ISP Replacement

Normally, Internet traffic passes through the ISP's routing infrastructure.

A VPN replaces the default routing path by directing traffic through VPN servers.

Benefits

- ISP cannot easily monitor browsing activities
- Improved anonymity
- Bypass geographical restrictions

Advantages of VPN

- Strong encryption
- Secure communication
- Protection on public Wi-Fi
- Location masking
- Privacy enhancement

VPN vs Proxy

Feature	Proxy	VPN
IP Hiding	Yes	Yes

Feature	Proxy	VPN
Encryption	No	Yes
Security	Basic	High
Privacy	Moderate	Strong
Speed	Usually Faster Slightly Slower	
Protects Entire Device	No	Yes

Importance in Cyber Security

Security professionals use proxies and VPNs for:

- Secure remote access
- Penetration testing
- Threat analysis
- Privacy protection
- Secure communications

Module 9: Web Technologies

Web Technologies refer to technologies used to create websites and web applications that users access through web browsers.

Understanding the difference between websites and web applications is important for web development, security testing, and cyber security.

23. Web Application vs Website

Although the terms Website and Web Application are often used interchangeably, they are different in functionality and complexity.

Web Application

A Web Application is a software application that runs through a web browser and allows users to interact with data and perform specific tasks.

Unlike websites, web applications are dynamic and interactive.

Examples

- Gmail
- Google Docs
- Facebook

- Instagram
- Amazon
- Online Banking Systems

Authentication

Most web applications require user authentication before granting access.

Authentication Methods

- Username and Password
- Multi-Factor Authentication (MFA)
- Biometric Authentication
- OTP Verification

Example

Username → Password → Login

Authentication verifies the user's identity.

Server-side Processing

Web applications perform significant processing on the server.

Examples

- User login validation
- Database operations
- Payment processing
- Order management

Example

User Request



Web Server



Database



Response

The server processes requests and returns results dynamically.

Characteristics of Web Applications

- Dynamic content

- User interaction
- Authentication required
- Database connectivity
- Business logic processing

Website

A Website is a collection of related web pages containing information accessible through a browser.

Websites primarily provide information rather than complex functionality.

Examples

- Company Websites
- Personal Blogs
- News Portals
- Educational Websites

Static Information

Many websites display static content that does not change frequently.

Examples

- Company information
- Contact details
- Product descriptions
- News articles

Users can view content but usually cannot perform complex operations.

Web Pages

A website consists of multiple web pages linked together.

Examples

Home Page
About Us
Services
Contact Us

Each page contains specific information.

Characteristics of Websites

- Information-focused

- Minimal user interaction
- Often static content
- Simpler architecture
- Faster development

Website vs Web Application

Feature	Website	Web Application
Purpose	Information Sharing	Task Execution
User Interaction	Limited	Extensive
Authentication	Usually Not Required	Usually Required
Database Usage	Minimal	Extensive
Content	Mostly Static	Dynamic
Examples	Blog, Company Site	Gmail, Facebook, Banking App

Importance in Cyber Security

Understanding web applications and websites helps security professionals:

- Identify security vulnerabilities
- Perform web application testing
- Analyze authentication mechanisms
- Detect insecure configurations
- Conduct penetration testing

Module 10: Authentication & Sessions

Authentication and Session Management are fundamental concepts in Web Security and Cyber Security. They ensure that users are properly identified, granted appropriate access, and can securely interact with web applications.

Almost every modern application such as Gmail, Facebook, Instagram, Banking Applications, and E-commerce platforms relies on authentication, authorization, sessions, and cookies to manage user access and maintain security.

24. Authentication

Authentication is the process of verifying the identity of a user, device, or application before granting access to a system.

In simple terms:

Authentication answers the question:

"Who are you?"

Before accessing a protected resource, the system must confirm that the user is genuinely who they claim to be.

Identity Verification

Identity verification is the process of confirming a user's identity using credentials or authentication factors.

Common Authentication Factors

Something You Know

- Password
- PIN
- Security Questions

Something You Have

- Mobile Phone
- Smart Card
- OTP Token

Something You Are

- Fingerprint
- Face Recognition
- Retina Scan

Example

Username: admin

Password: Password@123

The system verifies the credentials and confirms the user's identity.

User Validation

User validation ensures that the credentials entered by the user match the information stored in the system.

Authentication Process

User Login



Enter Username & Password



Server Checks Database



Valid Credentials?



Access Granted / Access Denied

Real-World Examples

- Gmail Login
- Facebook Login
- Online Banking Login
- Company Employee Portals

Types of Authentication

Single-Factor Authentication (SFA)

Uses only one authentication factor.

Example:

Username + Password

Multi-Factor Authentication (MFA)

Uses multiple authentication factors.

Example:

Password + OTP

Benefits:

- Improved security
- Reduced account compromise risk
- Protection against stolen passwords

Importance of Authentication

Authentication helps:

- Verify user identity
- Prevent unauthorized access
- Protect sensitive information
- Secure user accounts

25. Authorization

Authorization is the process of determining what actions an authenticated user is allowed to perform.

In simple terms:

Authorization answers the question:

"What are you allowed to do?"

Authorization always occurs **after authentication**.

Access Control

Access control determines which resources a user can access.

Example

In a company:

Admin



Can create users

Can delete users

Can manage settings

Employee



Can only view personal information

Even though both users are authenticated, their access rights are different.

User Permissions

Permissions define specific actions a user can perform.

Examples

Permission Description

Read	View data
Write	Create or modify data
Update	Edit existing records
Delete	Remove records
Execute	Run programs or actions

Role-Based Access Control (RBAC)

Many applications use roles to manage permissions.

Example

Role	Permissions
Admin	Full Access
Manager	Limited Administrative Access
User	Basic Access
Guest	View Only

Authentication vs Authorization

Authentication	Authorization
Verifies identity	Grants permissions
Answers "Who are you?"	Answers "What can you do?"
Happens first	Happens after authentication
Uses credentials	Uses roles and permissions

Example

Login Successful
↓
Authentication Complete
↓
Check User Role
↓
Authorization Applied

26. Session

A Session is a temporary interaction between a user and a web application after successful authentication.

Once a user logs in, the server creates a session to remember the user during future requests.

Without sessions, users would need to log in again on every page.

User Session Management

Session Management is the process of creating, maintaining, and terminating user sessions securely.

Session Lifecycle

Step 1: Login

User enters credentials.

Username + Password

Step 2: Session Creation

The server creates a unique Session ID.

Example:

Session ID:

A7F9B23D45XYZ

Step 3: Session Storage

The Session ID is sent to the browser.

Step 4: Session Validation

Every future request includes the Session ID.

User Request



Session ID Attached



Server Verifies Session

Step 5: Logout

Session is destroyed when the user logs out.

Benefits of Sessions

- Maintains login state
- Improves user experience
- Reduces repeated authentication
- Supports secure communication

Session Security Risks

Session Hijacking

An attacker steals a valid Session ID and impersonates the user.

Session Fixation

An attacker forces a victim to use a known Session ID.

Session Timeout

Sessions should automatically expire after inactivity.

27. Cookies

Cookies are small pieces of data stored by a web browser on a user's device.

They help websites remember users, preferences, and session information.

Browser Storage

Cookies are stored locally inside the browser.

Information Stored

- Session IDs
- User Preferences
- Login Information
- Shopping Cart Data
- Language Settings

Example

Cookie:

username=admin

theme=dark

language=en

Session Tracking

Cookies are commonly used to maintain user sessions.

Example

After login:

Server Creates Session



Session ID Stored in Cookie



Browser Sends Cookie



Server Recognizes User

Without cookies, websites would not remember users between requests.

Types of Cookies

Session Cookies

- Temporary
- Deleted when browser closes
- Used for active sessions

Persistent Cookies

- Stored for a specific duration
- Remain after browser closes
- Used for preferences and personalization

Secure Cookies

- Sent only over HTTPS
- Protect sensitive data

HttpOnly Cookies

- Cannot be accessed by JavaScript
- Helps prevent XSS attacks

Advantages of Cookies

- Maintain user sessions
- Store preferences
- Improve user experience
- Enable personalization

Cookie Security Risks

Cookie Theft

Attackers steal cookies to impersonate users.

Cross-Site Scripting (XSS)

Malicious scripts may access cookies.

Session Hijacking

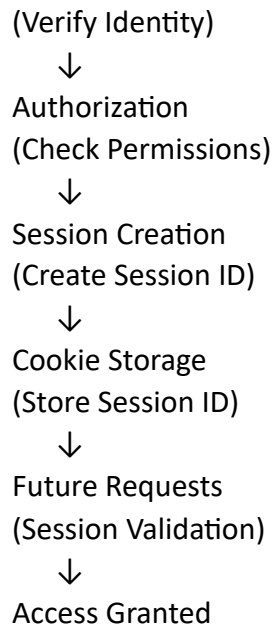
Stolen session cookies can provide unauthorized access.

Authentication, Authorization, Session & Cookies Workflow

User Login



Authentication



Importance in Cyber Security

Understanding Authentication, Authorization, Sessions, and Cookies helps security professionals:

- Secure user accounts
- Prevent unauthorized access
- Detect session hijacking attacks
- Implement secure authentication mechanisms
- Configure access controls
- Protect web applications

Module 11: Cyber Security Basics

Cyber Security is the practice of protecting computers, networks, servers, applications, and data from cyber threats, unauthorized access, and malicious attacks. As organizations increasingly rely on digital technologies, cyber security plays a critical role in ensuring the confidentiality, integrity, and availability of information.

Cyber Security helps individuals and organizations protect sensitive information, maintain business continuity, and prevent financial losses caused by cyber attacks.

28. What is Cyber Security?

Cyber Security refers to the technologies, processes, and practices designed to protect systems, networks, and data from cyber threats.

Objectives of Cyber Security

- Protect sensitive information

- Prevent unauthorized access
- Ensure system availability
- Maintain data integrity
- Detect and respond to threats

Information Security

Information Security focuses on protecting data from unauthorized access, disclosure, modification, or destruction.

Examples

- Protecting customer information
- Securing financial records
- Encrypting confidential documents

Importance

Information is one of the most valuable assets of an organization and must be protected throughout its lifecycle.

System Protection

System Protection involves safeguarding computers, servers, operating systems, and applications from attacks.

Methods

- Antivirus software
- Patch management
- Access controls
- Security monitoring

Goal

Ensure systems operate securely and efficiently.

Network Protection

Network Protection focuses on securing communication channels and network infrastructure.

Security Measures

- Firewalls
- Intrusion Detection Systems (IDS)
- Intrusion Prevention Systems (IPS)

- VPNs
- Network Monitoring

Goal

Prevent unauthorized access and malicious network activity.

29. Asset, Threat, Vulnerability and Risk

These four concepts form the foundation of risk management in cyber security.

Asset

An Asset is anything valuable that needs protection.

Examples

- Customer data
- Employee records
- Computers
- Servers
- Databases
- Intellectual property

Types of Assets

Physical Assets

- Computers
- Network devices
- Data centers

Digital Assets

- Databases
- Source code
- Documents

Human Assets

- Employees
- Customers
- Contractors

Threat

A Threat is anything capable of causing harm to an asset.

Examples

- Hackers
- Malware
- Insider attacks
- Natural disasters
- Hardware failures

Threat Sources

Internal Threats

Employees or contractors.

External Threats

Cybercriminals, attackers, or competitors.

Vulnerability

A Vulnerability is a weakness that can be exploited by a threat.

Examples

- Weak passwords
- Unpatched software
- Misconfigured servers
- Open network ports

Common Vulnerabilities

- SQL Injection
- Cross-Site Scripting (XSS)
- Default credentials
- Outdated software

Risk

Risk is the possibility that a threat will exploit a vulnerability and cause damage to an asset.

Examples

A company stores sensitive customer data.

- Asset → Customer Database

- Threat → Hacker
- Vulnerability → Weak Password
- Risk → Data Breach

Risk Formula

Risk can be understood using the following relationship:

Asset + Threat + Vulnerability = Risk

Example

Asset → Banking Database

Threat → Cyber Attacker

Vulnerability → Weak Authentication

Risk → Unauthorized Access

30. CIA Triad

The CIA Triad is the core security model used to design and evaluate information security systems.

CIA stands for:

- Confidentiality
- Integrity
- Availability

These three principles guide the implementation of security controls.

Confidentiality

Confidentiality ensures that information is accessible only to authorized individuals.

Methods

- Encryption
- Authentication
- Access Control
- Password Protection

Example

Only authorized employees can access payroll information.

Integrity

Integrity ensures that information remains accurate and unaltered.

Methods

- Hashing
- Checksums
- Digital Signatures
- Version Control

Example

A student's marks should not be modified by unauthorized users.

Availability

Availability ensures that systems and data remain accessible when needed.

Methods

- Backups
- Redundancy
- Load Balancing
- Disaster Recovery

Example

An online banking service should remain available 24/7.

CIA Triad Summary

Principle	Purpose
Confidentiality	Prevent unauthorized access
Integrity	Maintain data accuracy
Availability	Ensure continuous access

Module 12: Social Engineering

Social Engineering is a psychological manipulation technique used by attackers to trick individuals into revealing sensitive information or performing actions that compromise security.

Rather than attacking systems directly, social engineering attacks target human behavior.

31. Phishing

Phishing is a cyber attack where attackers send fraudulent messages pretending to be legitimate organizations.

The goal is usually to steal credentials, financial information, or personal data.

Fraudulent Emails

Attackers create emails that appear to come from trusted sources.

Examples

- Fake bank emails
- Fake PayPal notifications
- Fake Microsoft alerts

Typical Message

Your account has been suspended.
Click here to verify your identity.

Credential Theft

Attackers redirect victims to fake login pages.

Goal

Steal:

- Usernames
- Passwords
- Banking details
- Credit card information

Example

A fake Gmail login page designed to capture credentials.

Signs of Phishing

- Suspicious links
- Urgent language
- Poor grammar
- Unknown senders
- Requests for sensitive information

32. Vishing

Vishing stands for Voice Phishing.

It is a social engineering attack conducted through phone calls.

Voice Phishing

Attackers pretend to be:

- Bank representatives
- Technical support staff
- Government officials

Objective

Convince victims to reveal sensitive information.

Phone Scams

Examples include:

- OTP requests
- Bank account verification scams
- Credit card fraud calls

Example

Hello Sir,
Your bank account will be blocked.
Please provide your OTP immediately.

Module 13: Security Professionals

Cyber Security professionals play an important role in identifying vulnerabilities, testing security controls, and protecting systems against attacks.

33. Penetration Testers

A Penetration Tester (Ethical Hacker) is a security professional who legally tests systems and applications to identify vulnerabilities.

Ethical Hacking

Ethical Hackers use hacking techniques with authorization.

Responsibilities

- Find vulnerabilities
- Assess risks
- Recommend fixes
- Improve security posture

Security Testing

Penetration testers evaluate:

- Web applications
- Networks
- Mobile applications
- APIs
- Cloud environments

Goal

Identify weaknesses before attackers exploit them.

Bug Bounty Hunting

Organizations reward researchers for reporting vulnerabilities.

Examples

- Google Bug Bounty
- Microsoft Bug Bounty
- Meta Bug Bounty

Benefits

- Encourages responsible disclosure
- Improves security

34. Vulnerability Assessment and Penetration Testing (VAPT)

VAPT is a combination of Vulnerability Assessment and Penetration Testing used to evaluate security posture.

Vulnerability Assessment (VA)

Vulnerability Assessment identifies security weaknesses in systems and applications.

Activities

- Scanning systems
- Identifying vulnerabilities
- Risk classification
- Reporting findings

Tools

- Nessus
- OpenVAS
- Qualys

Penetration Testing (PT)

Penetration Testing goes beyond identification and attempts to exploit vulnerabilities.

Activities

- Exploitation
- Privilege escalation
- Security validation

Goal

Determine the real-world impact of vulnerabilities.

VA vs PT

Vulnerability Assessment Penetration Testing

Finds vulnerabilities	Exploits vulnerabilities
Automated scanning	Manual testing
Identifies weaknesses	Demonstrates impact
Broad coverage	Deep analysis

Module 14: Hackers and Cyber Attacks

Hackers and cyber attacks are major concerns for organizations and individuals worldwide.

Understanding attacker behavior helps security teams build effective defenses.

35. Hackers

A Hacker is a person who uses technical skills to access, analyze, or manipulate computer systems.

Ethical Hackers

Ethical Hackers work legally to improve security.

Responsibilities

- Security testing
- Vulnerability assessment
- Penetration testing
- Security consulting

Unauthorized Access

Malicious hackers attempt to gain unauthorized access to systems.

Objectives

- Data theft
- Financial gain
- Espionage
- Service disruption

Types of Hackers

White Hat Hackers

Authorized security professionals.

Black Hat Hackers

Malicious attackers.

Gray Hat Hackers

Operate between ethical and unethical boundaries.

Script Kiddies

Use existing tools without deep technical knowledge.

Hacktivists

Conduct attacks for political or social causes.

36. Types of Cyber Attacks

Cyber attacks are deliberate attempts to compromise systems, networks, or data.

36.1 Malware

Malware (Malicious Software) is software designed to damage, disrupt, or gain unauthorized access to systems.

Virus

A Virus attaches itself to legitimate files and spreads when executed.

Effects

- File corruption
- System slowdown
- Data loss

Trojan

A Trojan disguises itself as legitimate software.

Purpose

- Install malware
- Create backdoors
- Steal information

Worm

A Worm spreads automatically without user interaction.

Characteristics

- Self-replicating
- Rapid propagation
- Network congestion

Ransomware

Ransomware encrypts files and demands payment for decryption.

Impact

- Data loss
- Business disruption
- Financial damage

Spyware

Spyware secretly collects user information.

Targets

- Passwords
- Browsing history
- Personal information

Adware

Adware displays unwanted advertisements.

Effects

- Browser disruption
- Reduced performance

Backdoor

A Backdoor provides unauthorized access to a system.

Risk

Attackers can bypass authentication mechanisms

Scareware

Scareware uses fear tactics to trick users.

Example

Your computer is infected!

Click here to clean it now.

36.2 Phishing Attacks

Spear Phishing

Targeted phishing attack against specific individuals or organizations.

Characteristics

- Personalized messages
- Higher success rate

Whaling

Targets high-profile individuals.

Examples

- CEOs
- Executives
- Directors

Pharming

Redirects users to fake websites by manipulating DNS settings.

Goal

Steal credentials and sensitive information.

36.3 Man-in-the-Middle (MITM) Attack

A MITM attack occurs when an attacker secretly intercepts communication between two parties.

Communication Flow

User ↔ Attacker ↔ Server

Objectives

- Data interception
- Credential theft
- Session hijacking

Common Causes

- Public Wi-Fi
- ARP Spoofing
- Rogue Access Points

36.4 Denial of Service (DoS) Attack

A Denial of Service attack attempts to make a system or service unavailable by overwhelming it with excessive traffic.

Goal

Prevent legitimate users from accessing services.

Effects

- Website downtime
- Service disruption
- Financial losses

Types

DoS (Single Attacker)

One system launches the attack.

DDoS (Distributed Denial of Service)

Multiple compromised systems attack simultaneously.

Importance in Cyber Security

Understanding these concepts helps professionals:

- Identify cyber threats
- Assess risks
- Protect organizational assets
- Conduct security assessments
- Investigate incidents
- Build secure systems